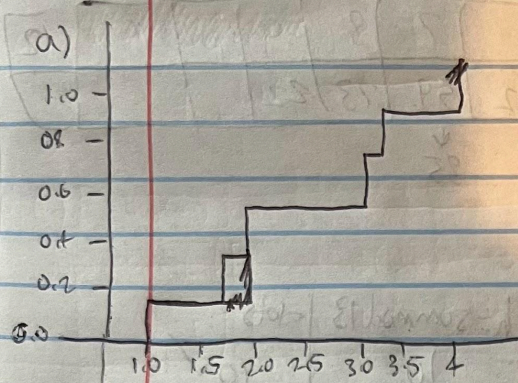
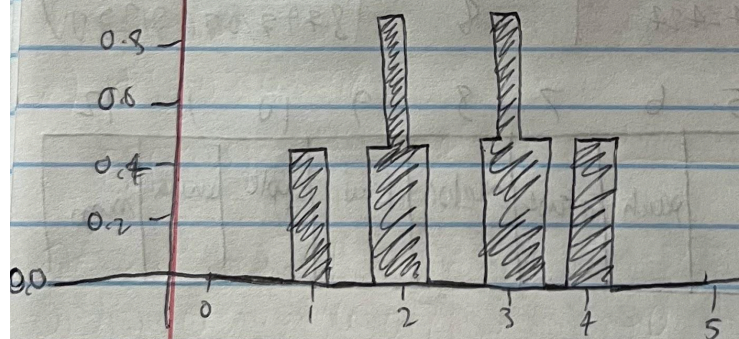


1) $[1, 1.75, 2, 3, 3.25, 4]$



b) Each point contributes a block of height $1/(2nh) = 1/(2 \cdot 6 \cdot 0.2) = 0.417$ on $(x_i - 0.2, x_i + 0.2)$, so,

- blocks 1 and 4 stand at height 0.417
- blocks around 1.75 and 2 overlap on $(1.8, 1.95)$, where height goes to around 0.833, same happens for $(3.05, 3.2)$
- everywhere else is 0



c) mean $\bar{x} = \frac{1 + 1.75 + 2 + 3 + 3.25 + 4}{6} = \frac{15}{6} = 2.5$

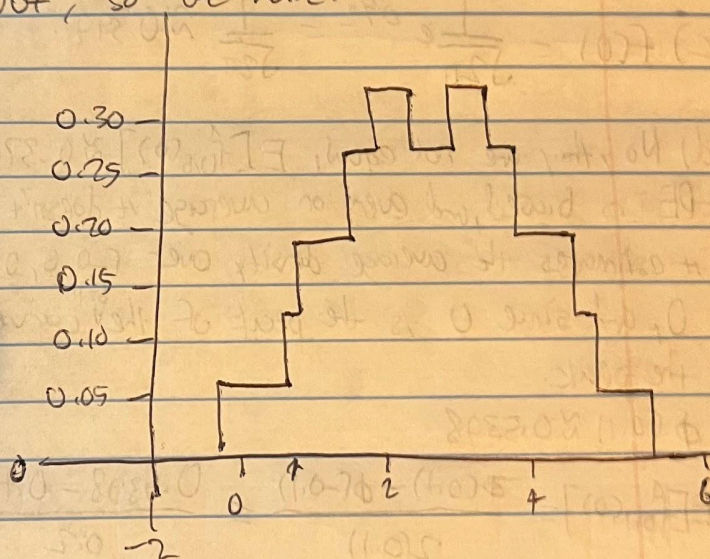
• SD: deviations $x_i - \bar{x}$ are: $-1.5, -0.75, -0.5, 0.5, 0.75, 1.5$, and their squares are: $2.25, 0.5625, 0.25, 0.25, 0.5625, 2.25$, and these sum to 6.125 , so:

$$SD = \sqrt{\frac{6.125}{6}} = \sqrt{1.0208} \approx 1.010$$

bandwidth: $6^{0.2} \approx 0.699 \rightarrow h_u = 1.84 \times 1.010 \cdot 0.699 \approx 1.30$

d) With $h=1.3$, each block has width $2h=2.6$ and height $\frac{1}{2.6 \cdot 1.30} = \frac{1}{3.38} \approx 0.064$, so we have:

point	Block $(x-1.3, x+1.3)$
1	$(-0.3, 2.30)$
1.75	$(0.45, 3.05)$
2	$(0.70, 3.30)$
3	$(1.70, 4.30)$
3.25	$(1.95, 4.55)$
4	$(2.70, 5.30)$



interval	# blocks	$f(x)$
$(-0.3, 0.45)$	1	0.064
$(0.45, 0.7)$	2	0.128
$(0.7, 1.70)$	3	0.192
$(1.70, 1.95)$	4	0.256
$(1.95, 2.3)$	5	0.321
$(2.3, 2.7)$	4	0.256
$(2.7, 3.05)$	5	0.321
$(3.05, 3.3)$	4	0.256
$(3.3, 4.3)$	3	0.192
$(4.3, 4.55)$	2	0.128
$(4.55, 5.30)$	1	0.064

e) Both (b) and (d) are uniform KDEs of the same data, both are step functions made of 'blocks', and both have total area 1, but in (b) $h=0.2$, which facilitates six more narrow spikes w/ no density in between. In (d), $h=1.30$ is wider, so the blocks overlap into a singular hump, which ~~suggests~~ suggests a density w/ a single peak and likely smoothed-away detail.

$$2) X = [-1, -0.5, 0, 0.5, 1]$$

a) ~~Three~~ Three of the above numbers are $< h=0.6$, so we have

$$\hat{f}_{0.6}(0) = \frac{3}{2 \cdot 5 \cdot 0.6} = \frac{3}{6} = 0.5$$

b) $E[\hat{f}_h(x)] = \frac{F(x+h) - F(x-h)}{2h}$. With $F = \Phi$, the standard Normal CDF, and $\Phi(0.6) \approx 0.7257$ via normal table;

$$E[\hat{f}_{0.6}(0)] = \frac{\Phi(0.6) - \Phi(-0.6)}{2(0.6)} = \frac{0.7257 - 0.2743}{1.2}$$

$$= \frac{0.4515}{1.2} \approx 0.376$$

$$c) f(0) = \frac{1}{\sqrt{2\pi}} e^{-0^2/2} = \frac{1}{\sqrt{2\pi}} \approx 0.399$$

d) No, they are not equal, $E[\hat{f}_{0.6}(0)] \approx 0.376 < f(0) \approx 0.399$; thus the KDE is biased, and even on average it doesn't equal the true density as it estimates the average density over $(-0.6, 0.6)$, but not the density at 0, and since 0 is the peak of the ^{bell} curve, that average is lower than the peak.

$$e) \Phi(0.1) \approx 0.5398$$

$$E[\hat{f}_{0.1}(0)] = \frac{\Phi(0.1) - \Phi(-0.1)}{2(0.1)} = \frac{0.5398 - 0.4602}{0.2} = \frac{0.0797}{0.2} \approx 0.3985$$

The gap shrinks from around $0.399 - 0.376 = 0.023$ to about $0.3989 - 0.3983 \approx 0.0007$, so as $h \rightarrow 0$, the secant $\frac{F(h) - F(-h)}{2h}$ approaches the derivative $F'(0) = f(0)$, so the bias goes to 0.

f) Yes, provided that the bandwidth shrinks as the sample grows. Indeed, w/ the Silverman plug-in, $h \propto n^{-1/5}$, thus $h_n \rightarrow 0$ as $n \rightarrow \infty$. From (e), the bias goes to 0 as $h \rightarrow 0$, so the KDE tends to the true density as n gets large, so it is biased but consistent.

g) 1) If it's too small, then few points fall in each window, thus the estimate would be noisy, 2) if it's too large, the estimate is over-smoothed, hiding features of the data. In general h should decrease w/ n for consistency.

3) $x = [5, 7, 7, 8, 10]$, $h = 1.5$
 • denominator is $2nh = 2 \cdot 5 \cdot 1.5 = 15$

x	Window $(x-1.5, x+1.5)$	points	$\hat{f}_h(x)$
6.0	(4.5, 7.5)	5, 7, 7	0.2
6.5	(5, 8)	5 7, 7	0.133
7.0	(5.5, 8.5)	7, 7, 8	0.2
8.0	(6.5, 9.5)	7, 7, 8	0.2
8.5	(7, 10)	8	0.067

b) $\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n \phi\left(\frac{x-x_i}{h}\right)$, where $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ and $nh = 5 \cdot 1.5 = 7.5$

x	z_i for 5, 7, 7, 8, 10	$\phi(z_i)$	sum	$\hat{f}_h(x) = \text{sum}/7.5$
6.0	0.667, -0.667, -0.667, -1.333, -2.667	0.3194, 0.3194, 0.3194, 0.1640, 0.0114	1.1338	0.151
6.5	1, -0.833, -0.833, -1, -2.333	0.2420, 0.3774, 0.3774, 0.2420, 0.0262	1.2699	0.169
7.0	1.333, 0, 0, -0.667, -2	0.1640, 0.3989, 0.3989, 0.3194, 0.0540	1.3353	0.178
8.0	2, 0.667, 0.667, 0, -1.333	0.0540, 0.3194, 0.3194, 0.3989, 0.1640	1.2558	0.167
8.5	2.333, 1, 1, 0.833, -1	0.0262, 0.2420, 0.2420, 0.3774, 0.2420	1.1795	0.151

c) Biggest jump is from $x=8.0$ to $x=8.5$, $0.200 \rightarrow 0.067$.

Here, the two 7's are exactly 1.5 away, so we don't count them, and thus the count is 1. Since the uniform kernel can be thought of as a box, a point would get full weight while it is less than h away and zero weight when once it reaches h , thus the estimate jumps whenever a data point reaches the edge of the window. Conversely, the gaussian kernel is a bell curve w/ no edges, so a point's weight decreases gradually as it moves further away.

d) Uniform is more simple and easy to interpret, as $\hat{f}$ is the share of points in a window around x divided by $2h$ - easier to compute by hand. But jumps depend on where exactly a point is relative to the window edges. Gaussian is smooth as each point contributes a little, so less sensitive to the window edges. Usually would give a more realistic view of continuous density, but can put some density beyond the actual data.

PROBLEM 4:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

[2] ✓ 2.6s Python

```
df = pd.read_csv('../data/heart_failure_clinical_records_dataset.csv')
df.head()
```

[3] ✓ 0.0s Python

...

	age	anaemia	creatinine_phosphokinase	diabetes	ejection_fraction	high_blood_pressure	platelets	serum_creatinine	serum_sodium	sex	smoking	time	DEATH_EVENT
0	75.0	0	582	0	20	1	265000.00	1.9	130	1	0	4	1
1	55.0	0	7861	0	38	0	263358.03	1.1	136	1	0	6	1
2	65.0	0	146	0	20	0	162000.00	1.3	129	1	1	7	1
3	50.0	1	111	0	20	0	210000.00	1.9	137	1	0	7	1
4	65.0	1	160	1	20	0	327000.00	2.7	116	0	0	8	1

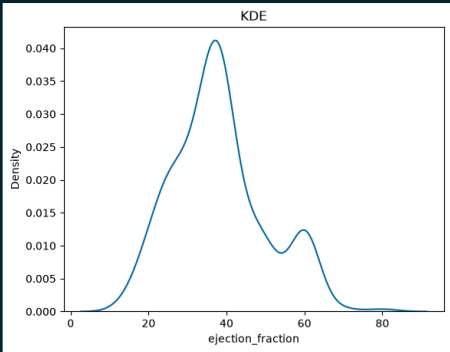
Part a)

• a. Plot a kernel density estimate of `ejection_fraction`.

```
sns.kdeplot(data=df, x='ejection_fraction').set(title='KDE')
plt.show()
```

[4] ✓ 0.1s Python

...



Part b)

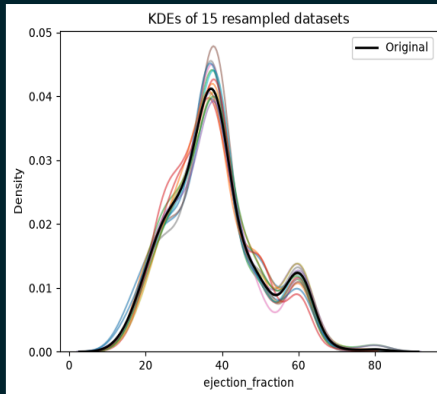
- b. Use Pandas' `df.sample(frac=1.0, replace=True)` to resample the data 15 times with replacement, plotting the kernel density estimates for each sample.

```
for s in range(15):
    df_s = df.sample(frac=1.0, replace=True) #resample w replacement
    sns.kdeplot(data = df_s, x = 'ejection_fraction', alpha = 0.5)

#plot original kde for reference
sns.kdeplot(data = df, x = 'ejection_fraction', color = 'black', linewidth = 2, label = 'Original')
plt.title('KDEs of 15 resampled datasets')
plt.legend()
plt.show()
```

[5] ✓ 0.3s

Python



Part c)

- c. For what values of `ejection_fraction` is the original plot reliable? For which values is there noticeable variation in your plots of the resampled data?

In terms of reliable values, looking at the above plots, it seems that around 15-35 and 40-50 are where the original plot is reliable, for the resampled curves sit close to on top of the black curve.

In terms of noticeable variation, areas such as the height of the main peak near 38, around 25-30, around 50-65, and somewhat in the tails below around 15 and above around 65.